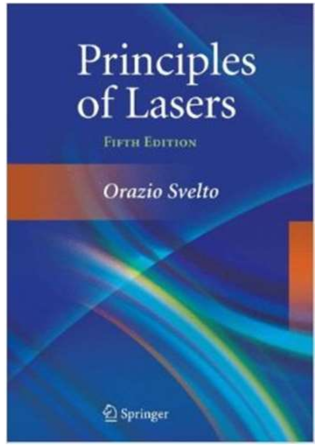
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《激光原理与技术》

Principles and Technologies of Lasers



Ch1-1
Basic Principles of Lasers

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Laser Fundamentals

... when the first lasers were operated, I and other scientists close to the research were surprised at how easy it turned out to be. We had assumed that, since lasers had never been made, it must be very difficult. But once you knew how, it was not at all difficult. Mostly what had been lacking were ideas and concepts.

- Arthur L Schawlow
- 1981 Nobel Prize for Laser Spectroscopy (Bertolotti, 1983)

1. Photon

Einstein wrote: " There are therefore now two theories of light, both indispensable, and … without any logical connection."

Evidence for wave-nature of light

- Diffraction and interference

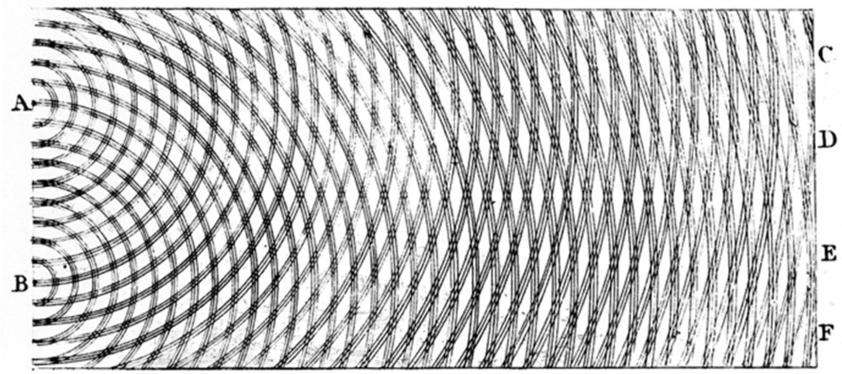
Evidence for particle-nature of light

- Photoelectric effect
- Compton effect

- ➢ Light demonstrates diffraction and interference phenomena that can only be explained in terms of its wave properties.
- ➢ Light is always detected as packets (photons); when observed, we never observe half a photon
- ➢ The number of photons is proportional to the energy density, which is related to the square of the electromagnetic field strength.

Wave-Particle Duality of Light

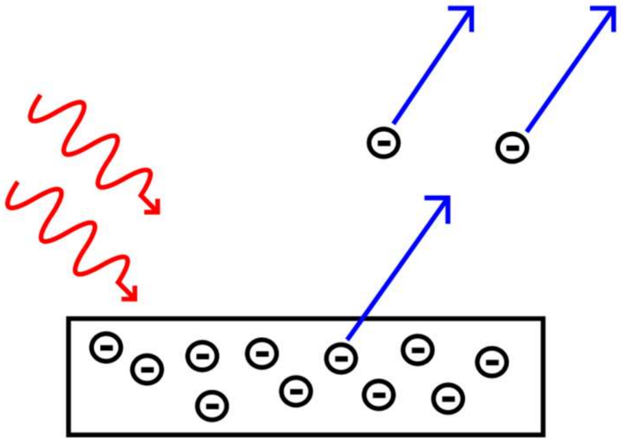
Evidence for wave-nature of light



Thomas Young's illustration of the two-slit experiment demonstrates the diffraction of light. His experiments provided strong support for the theory that light is composed of waves.

Evidence for particle-nature of light

A diagram illustrating the emission of electrons from a metal plate, where the energy of the incoming photon must exceed the material's work function for electron emission to occur.



The basic property of photon

1) The energy of photon
   $E=h\nu$, where h is the Planck constant $6.63*10^{-34}$ J·s

2) The mass of photon
   $m=E/C^2=\mathrm{h}\nu/C^2$ equal to zero with no movement

3) The momentum of photon P and the wave vector
   $\boldsymbol{P}=(mc)\boldsymbol{n}=(\mathrm{h}\nu/c)\boldsymbol{n}=(h/2\pi*2\pi/\lambda)\boldsymbol{n}=(h/2\pi)\boldsymbol{K}=\hbar\boldsymbol{K}$
   $\boldsymbol{K}=(2\pi/\lambda)\boldsymbol{n}$

4) Photon has two independent polarizations, corresponding to the polarization of the wave

5) Photon has spin, and the spin quantum number is an integer. Photon follows Bose-Einstein distribution and there is no limitation to the number of photons in the same state

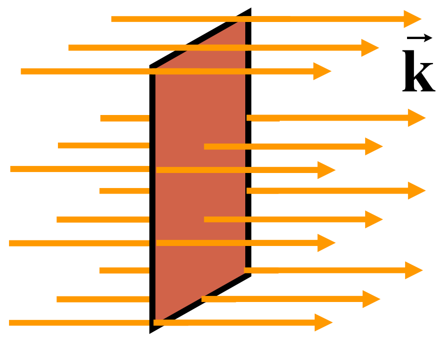
Mode & photon state

A photon is both a particle (with energy, mass, and momentum) and a wave (with frequency, wave vector, and polarization). These dual properties are well explained by Quantum Electrodynamics.

Wave-Nature: Arbitrary electromagnetic field can be viewed as a linear combination of plane waves with energy E and momentum $\boldsymbol{P}$, or as a superposition of eigenmodes (eigenstates) of the electromagnetic wave.

$$\vec{\mathbf{E}}(\mathbf{r},t) = \vec{\mathbf{E}}_0 e^{i(2\pi\nu t - \vec{\mathbf{k}}\cdot\vec{\mathbf{r}})}$$

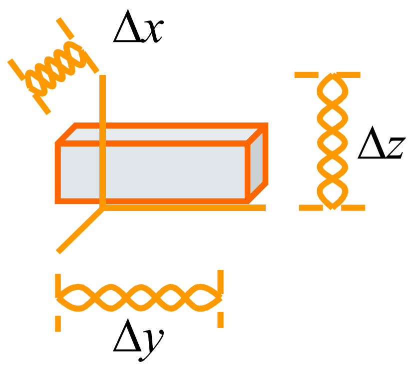
Free space, arbitrary $\boldsymbol{\nu}$ and $\boldsymbol{k}$



Only specific plane wave can resonate inside the cavity

$$V = \Delta x \Delta y \Delta z$$

$$\Delta x = m\frac{\lambda}{2} \qquad \Delta y = n\frac{\lambda}{2} \qquad \Delta z = q\frac{\lambda}{2}$$

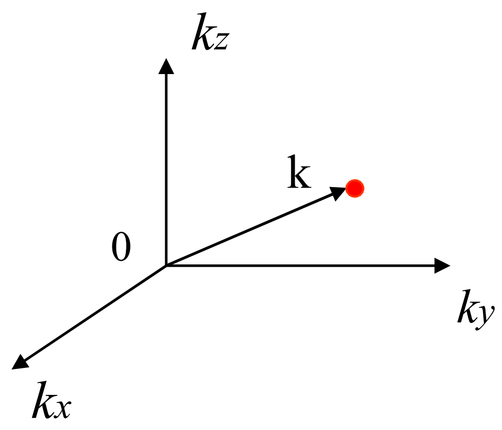


$$\left|\vec{k}\right| = \frac{2\pi}{\lambda} \Rightarrow \left|\vec{k}_x\right| = \frac{\pi}{\Delta x} m \qquad \left|\vec{k}_y\right| = \frac{\pi}{\Delta y} n \qquad \left|\vec{k}_z\right| = \frac{\pi}{\Delta z} q$$

Wave vector space

The space between two adjacent modes

$$\Delta k_x = \frac{\pi}{\Delta x} \qquad \Delta k_y = \frac{\pi}{\Delta y} \qquad \Delta k_z = \frac{\pi}{\Delta z}$$

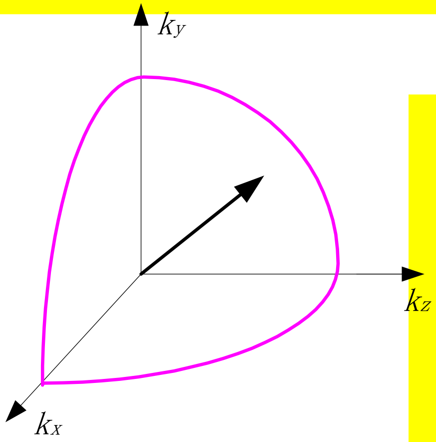


Mode density: the mode number inside unit volume and frequency

Unit volume $\Delta k_x \cdot \Delta k_y \cdot \Delta k_z = \dfrac{\pi^3}{\Delta x \Delta y \Delta z} = \pi^3 / V$ Wave vector space

for volume $|\vec{k}| \to |\vec{k}| + d|\vec{k}| \Rightarrow \left(\dfrac{1}{8}\right) 4\pi |\vec{k}|^2 d|\vec{k}|$

Mode number $\dfrac{1}{8} 4\pi |\vec{k}|^2 d|\vec{k}| \Big/ \left(\dfrac{\pi^3}{V}\right) = \dfrac{|\vec{k}|^2 d|\vec{k}| V}{2\pi^2} \Rightarrow \dfrac{4\pi\nu^2}{c^3} d\nu \cdot V$



$$N = 2 \times \frac{4\pi\nu^2}{c^3} V d\nu = \frac{8\pi\nu^2}{c^3} V d\nu$$

Two polarization

mode density $n_\nu = \dfrac{8\pi\nu^2}{c^3}$

Total mode number inside cavity: $N = \dfrac{8\pi v^2}{c^3} V dv$

Example: Calculate the mode number inside cavity $V = 2cm^3$

band width $\Delta\lambda = 1\times10^{-10}\,m$ $\lambda = 5\times10^{-7}\,m$

$$v = c/\lambda \qquad dv = \frac{c}{\lambda^2} d\lambda = \frac{\left(3\times10^{8}\right)\times\left(1\times10^{-10}\right)}{\left(5\times10^{-7}\right)^2} = 1.2\times10^{11}\,(\text{Hz})$$

$$N = \frac{8\pi v^2}{c^3} V dv = \frac{8\pi}{\lambda^2 c} V dv$$

$$= \frac{8\pi}{\left(5\times10^{-7}\right)^2 \times 3\times10^{8}} 2\times10^{-6} \times 1.2\times10^{11} = 8\times10^{10}$$

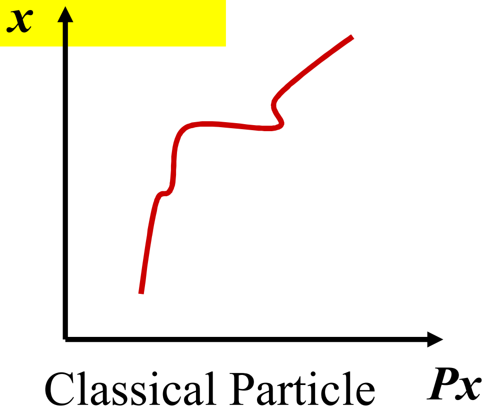
PS: For microwave, the mode density is quite low inside the 3D cavity. This is a big challenge for visible light

Close cavity ⇨ Open cavity

Phase space

Particle-Nature: Heisenberg's uncertainty principle

$$\Delta x \Delta P = h$$



Particle can not be distinguished inside the 2D phase space, they are belong to the same state

Similarly, for 3D movement

$$\Delta x \Delta y \Delta z \Delta P_x \Delta P_y \Delta P_z = h^3$$

The above equation represents the unit volume of the photon state in phase space, which is the smallest distinguishable unit. The movement of a photon is confined to a unit area in phase space, but its position cannot be precisely fixed. Therefore, the motion of photons in phase space is not continuous.

Unit volume $\Delta k_x \cdot \Delta k_y \cdot \Delta k_z = \dfrac{\pi^3}{\Delta x \Delta y \Delta z} = \pi^3 / V$ **Wave vector space**

$$\Delta P_x = 2\hbar \Delta k_x, \Delta P_y = 2\hbar \Delta k_y, \Delta P_z = 2\hbar \Delta k_z$$

$$\Delta P_x \Delta P_y \Delta P_z = 8\hbar^3 \Delta k_x \Delta k_y \Delta k_z = 8\hbar^3 \frac{\pi^3}{\Delta x \Delta y \Delta z} = \frac{h^3}{\Delta x \Delta y \Delta z}$$

$$\boxed{\Delta P_x \Delta P_y \Delta P_z \Delta x \Delta y \Delta z = h^3}$$

Conclusion

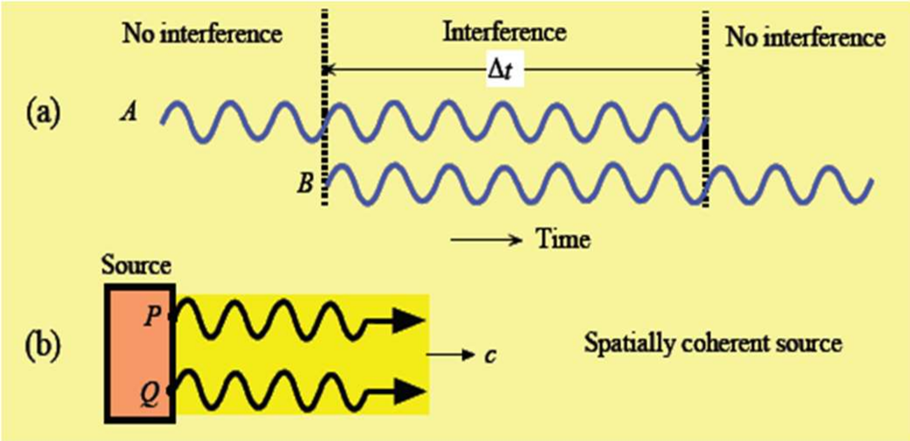
- One light mode has one position in phase space
- The mode of light = the state of photon
- They have the same space volume

$$\Delta x \Delta y \Delta z = \frac{h^3}{\Delta P_x \Delta P_y \Delta P_z}$$

Coherence of photon

time

space

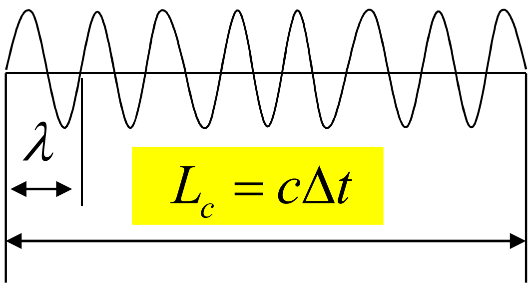


Coherence volume, area, length, and time

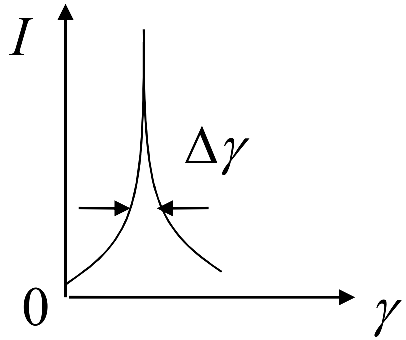
$$V_c = A_c L_c$$

$$V_c = A_c \tau_c c$$

$$\tau_c = L_c / c$$



$$\Delta\gamma = \frac{1}{\Delta t}$$



$$\tau_c = \Delta t = \frac{1}{\Delta\gamma}$$

Conclusion: Better monochromatic, longer coherence time

2. Quantum Physics

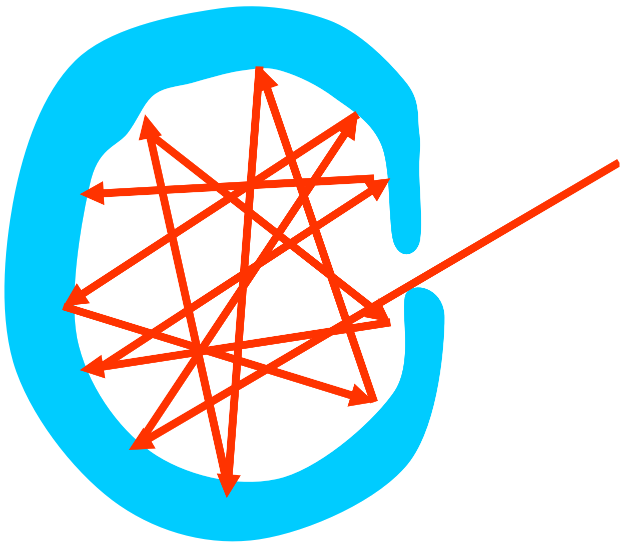
❖ Black body radiation

"Blackbody radiation" or "cavity radiation" refers to an object or system which absorbs all radiation incident upon it and re-radiates energy which is characteristic of this radiating system only, not dependent upon the type of radiation which is incident upon it. The radiated energy can be considered to be produced by standing wave or resonant modes of the cavity which is radiating

1) Heat radiation
2) Black body

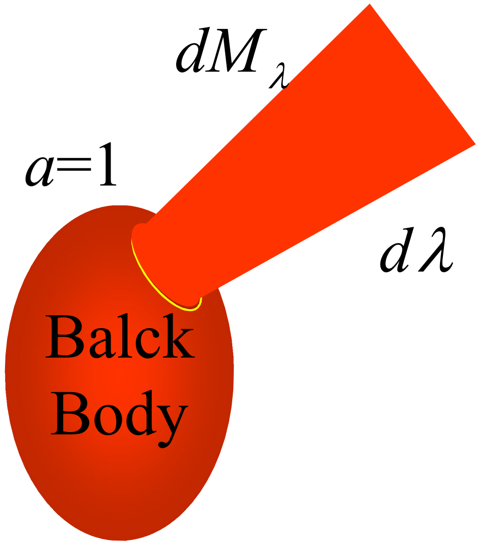
Absorption coefficient (a):
Mirror 0.02; Metal 0.6-0.8; Black particle: 0.95

Let us construct cavity in a metal block, through the wall of which a small hole is drilled. This small hole is called a black body. (a=1)



a) Unit spectral radiancy（单色辐出度）

The quantity $M_{(\lambda T)}$ is the rate at which energy is radiated per unit time, per unit wavelength and per unit area of surface for wavelengths lying in the interval λ to $\lambda+d\lambda$ at the temperature T.



$$M_{(\lambda.T)} = \frac{dM_{\lambda}}{d\lambda} \qquad (\mathrm{W}m^{-3})$$

It is a function of wavelength and time.

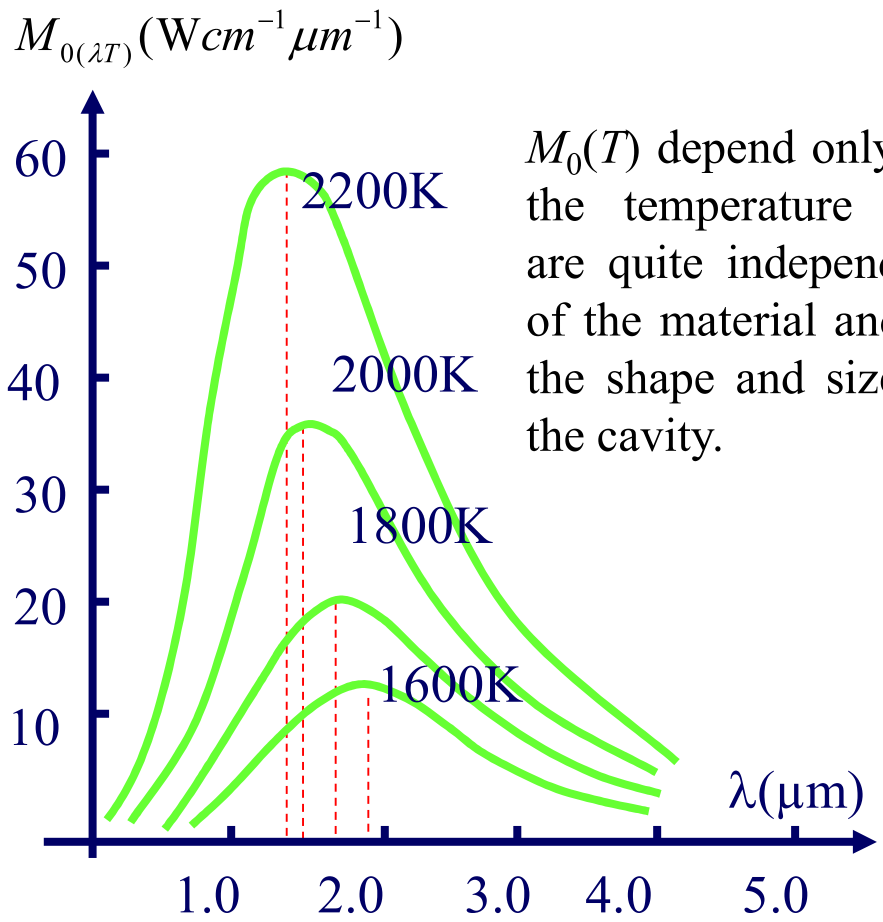
b) Spectral radiancy（辐出度）

The appropriate quantity $M_{(T)}$ is the rate at which energy is radiated per unit time and per unit area of surface at the temperature T.

$$M_{(T)} = \int_0^{\infty} M_{(\lambda.T)} d\lambda$$

The law about black-body radiation

Characteristics of Blackbody Radiation



$M_0(T)$ depend only on the temperature and are quite independent of the material and of the shape and size of the cavity.

a) J.Stefan-Ludwig Boltzmarn Law

$$M_{0(T)} = \sigma T^4$$

$$\sigma = 5.67 \times 10^{-8}\,\mathrm{W} / mK$$

J.Stefan-Ludwig Boltzmarn constant

b) Wien Displacement Law

$$T\lambda_m = b$$

$$b = 2.898 \times 10^{-3}\, mK$$

λ_m : wavelength for peak value

The law about black-body radiation

A number of capable physicists advanced theories based on classical physics, which, however, had only limited success.

Example 2:

Rayleigh-jean’s formula

$$M_{0(\lambda T)} = \frac{2\pi CKT}{\lambda^4}$$

C speed of light

K=1.380658×10^{-23}J/K

Boltzmann’s constant

Rayleigh-jean’s Formula isn’t fit the experimental points at short wavelength.

Ultraviolet Catastrophe

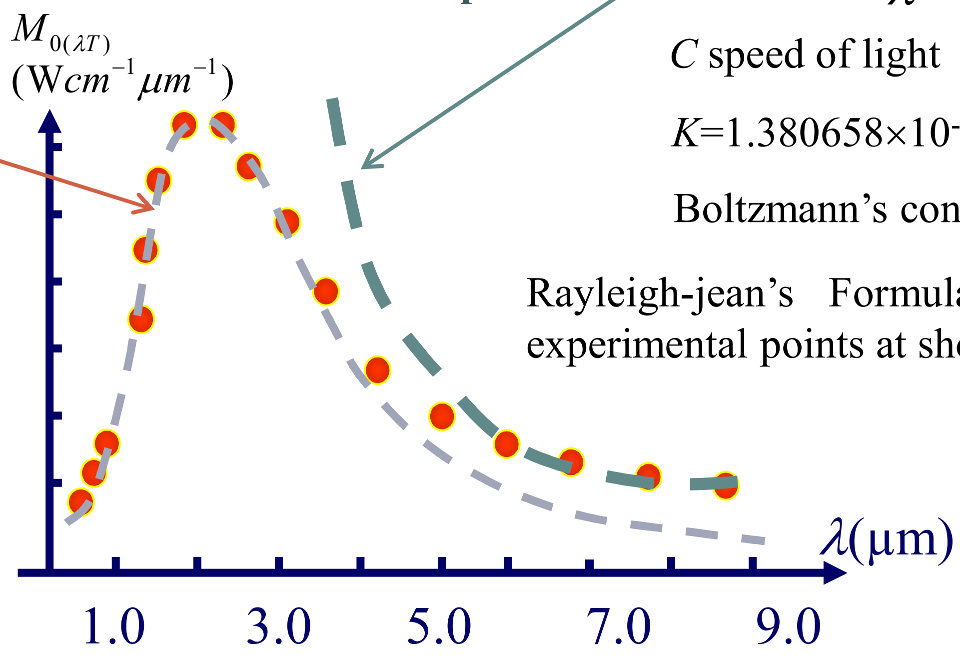
Example 1:

1896, Wien’s formula

$$M_{0(\lambda T)} = \frac{c_1}{\lambda^5} e^{-\frac{c_2}{\lambda T}}$$

Where C_1 and C_2 are constants that must be determined by the experimental data.

Wien’s formula isn’t fit to the experimental points at long wavelength.



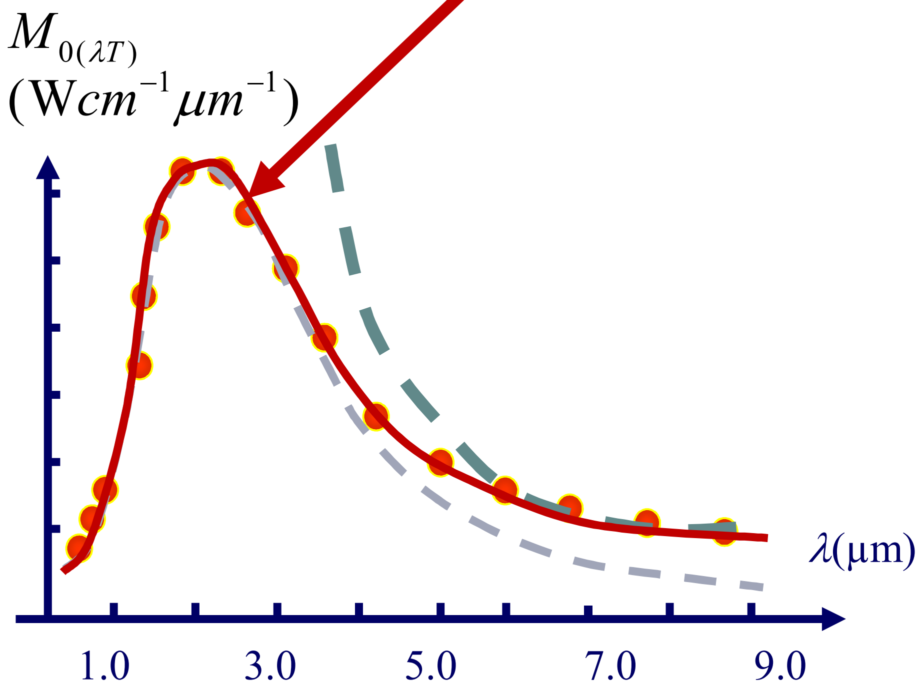
Planck's formula

In 1900 Max Planck pointed out that if Wien's formula were modified in a simple way it would prove to fit the data precisely. It is

$$M_{0(\lambda T)} = 2\pi hC^2\lambda^{-5}\frac{1}{e^{\frac{hC}{k\lambda T}}-1} = 2\pi hC^2\lambda^{-5}\frac{1}{e^{\frac{h\nu}{kT}}-1}$$

k: Boltzmann's constant

h=6.63×10^{-34}(Js) is called Planck's constant.



Planck's formula fit the data precisely.

Planck's Quantum Supposition

The atoms that make up blackbody walls behave like tiny electromagnetic oscillators. An oscillators can't have any energy but only energies given by

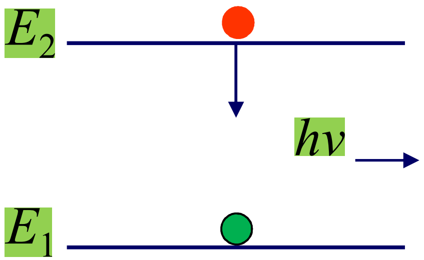
$$E_n = n\varepsilon \qquad n = 1.2.3\cdots$$

where *n* is a quantum number,

$$\varepsilon = h\nu \qquad h = 6.63\times10^{-34}\ js$$

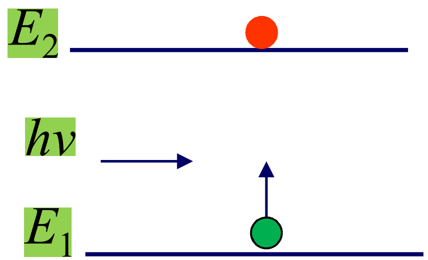
Interaction between light & matter

❖ Basic processes (A. Einstein, 1916)



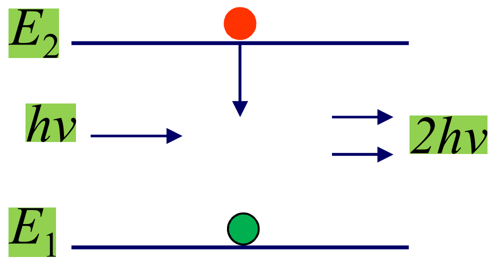
自发辐射(SP)
Spontaneous Emission

- ✓ Transition probability A_{21}
- ✓ Random phase and direction



受激吸收(STA)
Stimulated Absorption

- ✓ Transition probability W_{12}
- ✓ $\nu = E_2 - E_1 / h$



受激辐射(STE)
Stimulated Emission

- ✓ Transition probability W_{21}
- ✓ Has the same frequency and phase as the incident light

$\Rightarrow$ light amplification

ρ_ν —Radiation field (J m^{-3} s)

辐射场

$$\rho_\nu = \frac{8\pi\nu^2}{c^3}\frac{h\nu}{e^{\frac{h\nu}{kT}}-1}$$

SP probability A_{21}

$$A_{21} = \left(\frac{dn_{21}}{dt}\right)_{sp}\frac{1}{n_2} \qquad A_{21} = 1/\tau_s$$

STA probability W_{12}

$$W_{12} = \left(\frac{dn_{12}}{dt}\right)_{st}\frac{1}{n_1} \qquad W_{12} = B_{12}\rho_\nu$$

STE probability W_{21}

$$W_{21} = \left(\frac{dn_{21}}{dt}\right)_{st}\frac{1}{n_2} \qquad W_{21} = B_{21}\rho_\nu$$

A_{21}, W_{12} and W_{21} related to the property of atom

Relation between A_{21} & τ_s

$$-\frac{dn_2}{dt} = \left(\frac{dn_{21}}{dt}\right) = -\frac{n_2}{\tau_s} \Rightarrow n_2 = n_{20} e^{-t/\tau_s}$$

$$A_{21} = \left(\frac{dn_{21}}{dt}\right)_{sp} \frac{1}{n_2}$$

$$-\frac{dn_2}{dt} = \left(\frac{dn_{21}}{dt}\right)_{sp} = -A_{21} n_2 \Rightarrow n_2 = n_{20} e^{-A_{21} t}$$

$$\left.\begin{array}{l} n_2 = n_{20} e^{-t/\tau_s} \\ n_2 = n_{20} e^{-A_{21} t} \end{array}\right\} A_{21} = 1/\tau_s$$

- ❖ The number of particles on energy level E_2 shows exponential decay with time
- ❖ The probability of spontaneous emission is reverse to the lifetime of the energy level (time can be measured)
- ❖ A_{21} is only determined by the property of the materials, not related to the radiation field

Einstein Relations - A and B Coefficients

Black-body radiation
$$\rho_\nu = \left(\frac{8\pi\nu^2}{c^3}\right)\left(\frac{h\nu}{e^{h\nu/kT}-1}\right)$$

$$\left(\frac{dn_{21}}{dt}\right)_{sp} + \left(\frac{dn_{21}}{dt}\right)_{st} = \left(\frac{dn_{12}}{dt}\right)_{st}$$

$$A_{21}n_2 + B_{21}\rho_\nu n_2 = B_{12}\rho_\nu n_1$$

Boltzmann Distribution
$$\frac{n_2}{n_1} = \frac{f_2}{f_1}\exp\left(-\frac{E_2 - E_1}{kT}\right)$$

$$\frac{c^3}{8\pi h\nu^3}\left(e^{h\nu/kT}-1\right) = \frac{B_{21}}{A_{21}}\left(\frac{B_{12}}{B_{21}}\frac{f_1}{f_2}e^{h\nu/kT}-1\right)$$

f1
f2代表什么要掌握

Einstein Relations

$$\frac{A_{21}}{B_{21}} = \frac{8\pi h\nu^3}{c^3} = n_\nu h\nu$$

$$B_{12}f_1 = B_{21}f_2$$

$f_1 = f_2$

$$B_{12} = B_{21}$$

$$W_{12} = W_{21}$$

Note

1. The 3 processes exist all the time, but their intensity is not the same

2. Thermal equilibrium: SP or STE dominate?

$$I_{sp} = n_2 A_{21} h\nu \qquad I_{ste} = n_2 B_{21} \rho_\nu h\nu$$

T=1500 K, l=500 nm cavity

$$\frac{I_{ste}}{I_{sp}} = \frac{B_{21}}{A_{21}} \rho_\nu = \frac{c^3}{8\pi h\nu^3} \rho_\nu = \frac{1}{e^{h\nu/kT} - 1} = 2 \times 10^{-9}$$

3. Thermal equilibrium: 2 stimulated processes which dominate? $n_2 < \frac{f_2}{f_1} n_1$

Unit time STE photon density increase $\quad W_{21} n_2 = B_{21} \rho_\nu n_2$

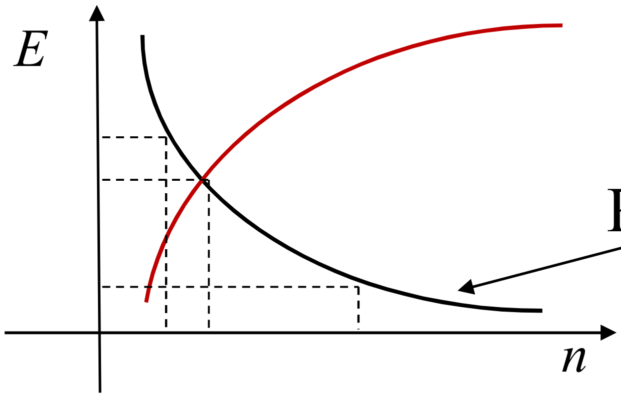
Unit time STA photon density decrease $\quad W_{12} n_1 = B_{12} \rho_\nu n_1$

4. The photon from spontaneous emission can be used as the out coming photon for stimulated absorption and stimulated emission

The Principle of Laser

激光三大要素：
pump
谐振腔
增益介质

1. Population Inversion



$n_2 > n_1$ (Population Inversion)

（粒子数反转是激光产生的客观条件）

RT, Bolzman distribution $\dfrac{n_2}{n_1} = e^{-\frac{(E_2 - E_1)}{kT}}$

- Material with population inversion is called gain/active material
- Population inversion is one of the requirement for lasing operation
- How to realize population inversion?

Energy is needed from outside to break the thermal equilibrium optically or electrically

The Principle of Laser

2. Light amplification

Spontaneous emission ~ not coherence light ~ low photon degeneracy
Stimulated emission ~ coherence light ~ high photon degeneracy

photon degeneracy $\Leftrightarrow$ The number of photon in the same state $\Leftrightarrow$ The number of photon with the same mode

The number of photon in the coherence volume $\Leftrightarrow$ The number of photon in the same unit phase space

1) Photon degeneracy of black body radiation $\bar{n} = \dfrac{E}{h\nu} = \dfrac{1}{e^{h\nu/kT} - 1}$

T=300 K: λ= 30 cm $\bar{n}$ =10^3; λ= 60 μm $\bar{n}$= 1; λ= 0.6 μm $\bar{n}$=10^{-35}

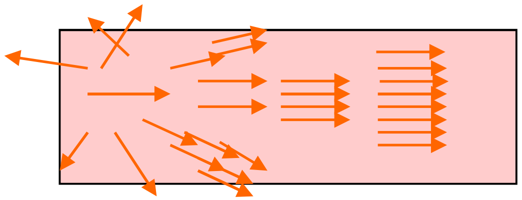
$$\rho_\nu = \left(\frac{8\pi\nu^2}{c^3}\right)\left(\frac{h\nu}{e^{h\nu/kT} - 1}\right) \Rightarrow \bar{n} = \frac{\rho_\nu}{\frac{8\pi h\nu^3}{c^3}} = \frac{B_{21}\rho_\nu}{A_{21}} = \frac{W_{21}}{A_{21}}$$

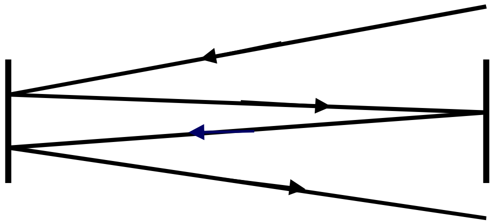
Photon degeneracy $\bar{n} = \dfrac{W_{21}}{A_{21}}$ =总光子数/模式数

To have lasing, it requires

$$\bar{n} >> 1 \rightarrow W_{21} \uparrow \rightarrow \rho_{\nu} \uparrow$$

- We need to increase the number of photon in the specific mode ρ_{ν}, and decrease the photon number of other modes to have coherence STE
- The photons are in one or more modes
- From close cavity to open cavity





- Open resonator only select part of the cavity mode, while other modes will escape from the cavity. Therefore the longitudinal mode has very high photon degeneracy
- Cavity has the function of mode selection, it is the main component of laser device

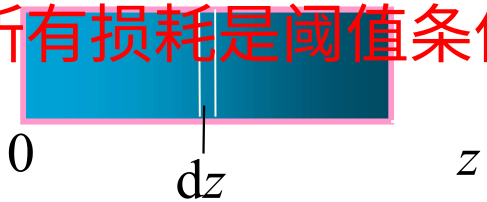
2) Gain Coefficient and the curve of gain

<u>Gain Coefficient</u>: the improved percentage of light pass through unit length of the gain material

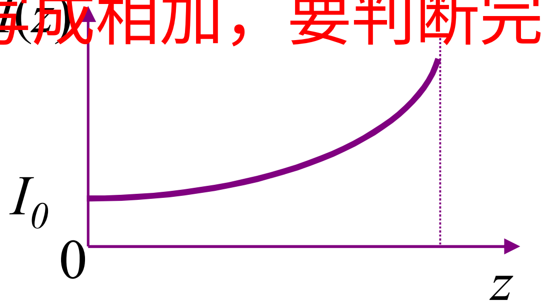
$$g(z)=\frac{dI(z)}{dz}\cdot\frac{1}{I(z)}$$

Gain material

增益=所有损耗是阈值条件



如果写成相加，要判断完不完整



net STE photon density

$$dI(z)\propto\left(\underline{W_{21}n_2-W_{12}n_1}\right)h\nu\, dz$$

$$W_{21}=B_{21}\rho_\nu \quad \Downarrow \quad f_1=f_2$$

$$dI(z)\propto B_{21}h\nu\rho(z)\left[n_2(z)-n_1(z)\right]dz$$

$$\Downarrow$$

$$g(z)\propto B_{21}h\nu\left[n_2(z)-n_1(z)\right]$$

Δ*n*(z)

$$\Downarrow$$

$$\boxed{g(z)\propto\Delta n(z)}$$

$$g(z) \propto B_{21} h\nu \Delta n(z) \quad \begin{cases} \Delta n(z) > 0 & g(z) > 0 & \text{Light amplified} \\ \Delta n(z) < 0 & g(z) < 0 & \text{Light absorbed} \\ \Delta n(z) = 0 & g(z) = 0 & \text{Transparent} \end{cases}$$

$$g(z) = \frac{dI(z)}{dz} \cdot \frac{1}{I(z)} \quad \Rightarrow \quad I(z) = I_0 e^{g(z)z}$$

If Δn is uniformly distributed, $g(z) \rightarrow$ constant g^0

<u>Weak signal gain coefficient</u>

$$I(z) = I_0 e^{g(z)z} \quad \Rightarrow \quad I(z) = I_0 e^{g^0 z}$$

With the increase of the light inside the material, whether the gain coefficient can remain unchanged?

$$I \uparrow \rightarrow \rho_\nu \uparrow \rightarrow \Delta n \downarrow \rightarrow g \downarrow$$

受激辐射消耗反转粒子数

Saturation of populated carriers, gain saturation

Gain Saturation $g(I)$

When the intensity of light increase to certain value, gain will decrease with the further increase of light intensity.

Q: When it happens?

Saturation intensity I_s: will be determined by the material itself (will be discussed in Ch_3)

$$\Delta n(z) = \frac{\Delta n}{1 + \frac{I(z)}{I_s}} \quad \xrightarrow{g \propto \Delta n} \quad g(I) = \frac{g^0}{1 + \frac{I(z)}{I_s}}$$

$$\begin{cases} I(z) << I_s & g(z) = g^0 \\ I(z) \sim I_s & g(I) \\ I(z) > I_s & g \downarrow \end{cases}$$

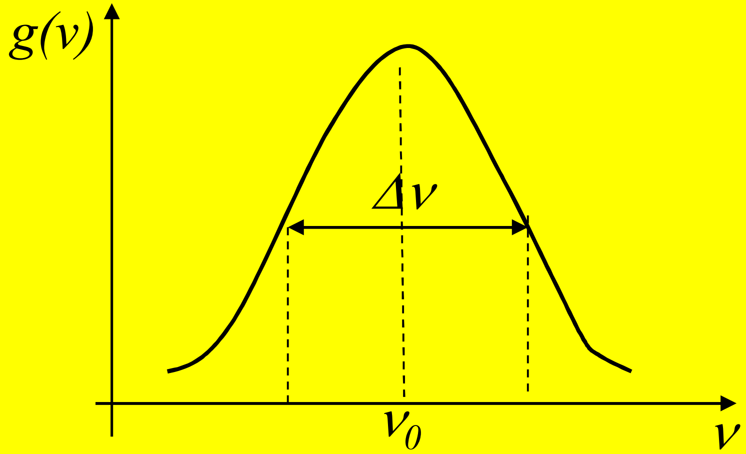
Weak signal gain coefficient, constant

Intensive signal gain coefficient, $g(I) < g^0$

Gain saturation

The curve of gain $g(\nu)$ — The gain coefficient $g(\nu)$ with frequency

Q: STE is not single frequency but has certain frequency distribution (Ch_3)



The width: $\Delta\nu$

Corresponds to the width of frequency at $\dfrac{g(\nu_0)}{2}$

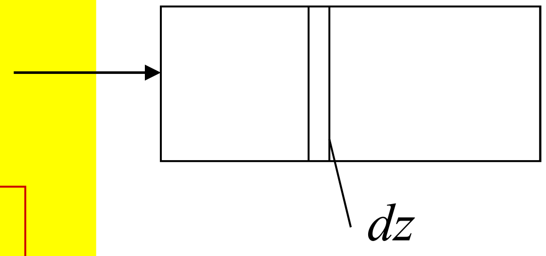
Loss coefficient a → negative gain coefficient $\alpha = -g$

$$\alpha(z) = -\frac{dI(z)}{dz} \cdot \frac{1}{I(z)} \Rightarrow \begin{cases} \alpha(z) \propto B_{21} h\nu \left[n_2(z) - n_1(z) \right] \\ I(z) = I_0 e^{-\alpha z} \end{cases}$$

g & a co-exist, then it change to

$$dI(z) = \left[g(I) - \alpha \right] I(z) dz$$

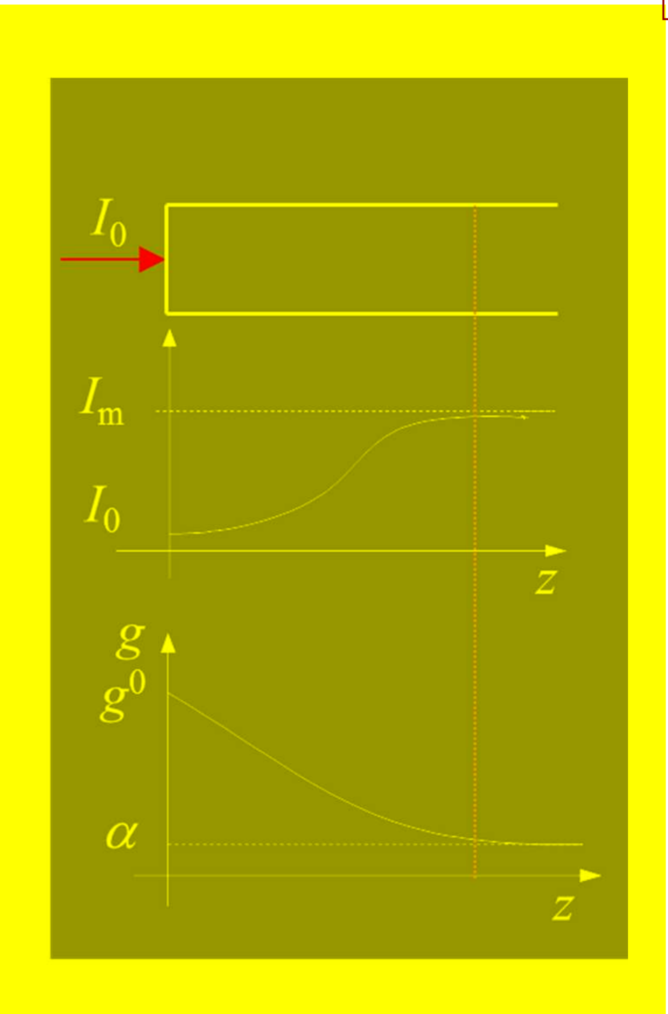
$$g(z) = g^0 \Rightarrow \boxed{I(z) = I_0 e^{(g^0 - \alpha) z}}$$



3) Self-oscillation

$$I(z) = I_0 e^{(g^0-\alpha)z}$$

$$I_0 \rightarrow \overset{I(z)\geq I_0}{I(z)\uparrow} \rightarrow g(z)\downarrow \xrightarrow{g(z)>\alpha} I(z)\uparrow \rightarrow g(z)\downarrow \xrightarrow[g(z)=\alpha]{} \boxed{I(z)=I_m}$$

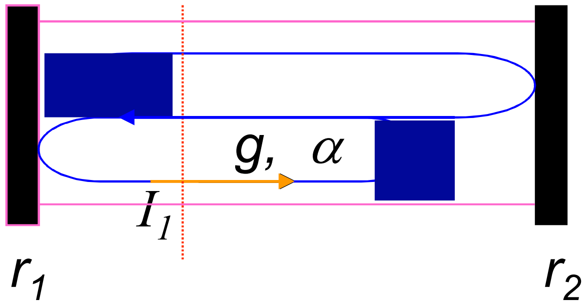


Only when the gain material is long enough, no matter how weak the light intensity I_0 at the first beginning, through gain saturation, finally can reach the maximum light intensity I_m, this is called self-oscillation.

- The oscillation is not related to the intensity of light at the very first beginning.
- The light comes from spontaneous emission
- Gain and gain saturation determines the final output power of the laser

How to fabricate gain material long enough?

Self-oscillation condition (threshold condition)



Matrix with refractive index *n* and length *L*

$$I_1 = I_0 e^{(g^0-\alpha)L} r_2 e^{(g^0-\alpha)L} r_1 = r_1 r_2 I_0 e^{2(g^0-\alpha)L} > I_0$$

$$r_1 r_2 e^{2(g^0-\alpha)L} \geq 1 \Rightarrow e^{2g^0L - 2\underbrace{\left(\frac{1}{2}\ln\frac{1}{r_1 r_2} + \alpha L\right)}_{=\delta}} \geq 1 \Rightarrow e^{g^0L-\delta} > 1$$

threshold condition

$$\boxed{g^0 L > \delta \quad \text{or} \quad g^0 > \alpha}$$

$$\delta = \frac{1}{2}\ln\frac{1}{r_1 r_2} + \alpha L$$

Single loop lost